

Mars Atmospheric Entry, Descent and Landing (EDL) Manual

1. Entry Interface Conditions

Entry trajectory begins at 125 km altitude with velocity of 6.1 km/sec. Heat shield thermal protection system utilizes PICA-X ablative tiles designed to endure heat fluxes exceeding 450 W/cm².

2. Supersonic Parachute Deployment

At Mach 2.2 and 10.5 km altitude, a 21.5-meter disk-gap-band parachute deploys under mortar ejection, shedding 92 percent of kinetic energy before radar ground acquisition.

3. Terminal Sky Crane Retrorockets

Terminal touchdown utilizes four throttleable hydrazine rocket clusters producing 3.1 kN each, descending at a steady 0.75 m/sec until laser contact sensors trigger engine cutoff.